Curriculum Vitae of M. Monirul Islam



Name : **M. Monirul Islam**

Position held : Professor, Dept. of Biochemistry & Molecular Biology, University of Chittagong since December 2015.

Present address : Dept. of Biochemistry & Molecular Biology, University of Chittagong, Chittagong-4331, Bangladesh.

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Academic Degrees

Subject	Degree	University	Country	Class	Year
Biophysics	Ph.D.	Tokyo University of Agriculture & Technology	Japan	Completed	2009
Biochemistry	M.Sc.	University of Dhaka	Bangladesh	1st class 1st	1998
Biochemistry	B.Sc.	University of Dhaka	Bangladesh	1st class 3rd	1997

Job Experiences

Institute	Location	Position	Duration
University of Chittagong	Bangladesh	Professor	Dec. 2015- till to date
University of Chittagong	Bangladesh	Associate Professor	Oct. 2009- Dec. 2015
University of Chittagong	Bangladesh	Assistant Professor	Jun. 2005- Oct. 2009
University of Chittagong	Bangladesh	Lecturer	Feb. 2003-Jun. 2005
Tokyo University of Agriculture & Technology	Japan	Invited Assistant Professor	Mar. 1- 31, 2005
ICDDR,B	Bangladesh	Research Officer	Aug. 1999-Feb. 2003

Research Experiences

Institute	Location	Position	Duration
Global Innovation Research	Japan	Guest Professor	Once in every year since 2017
JASSO	Japan	Follow-up Researcher	Nov. 2017-Jan. 2018
Japanese Govt. (JSPS)	Japan	Invitation Fellowship	Jun. 2014-Mar. 2015
Japanese Govt. (JSPS)	Japan	Post-doctoral fellow	Sep. 2009-Sep. 2011

Japanese Govt. (MEXT)	Japan	Ph.D. Research	Oct. 2005-Mar. 2009

Research Position/Fellowship/Experiences

JASSO Follow-up Research (November 2017-January 2018)

Institute : Tokyo University of Agriculture & Technology
Research Area/Specialization : Designing Chimeric dengue ED3s, their expression, purification and biophysical characterization

JSPS Invitation Research (June 2014-March 2015)

Institute : Tokyo University of Agriculture & Technology
Research Area/Specialization : Control of protein oligomerization and effects of sub-visible protein oligomer on the generation of serum antibody responses

Post-Doctoral Research (September 2009-September 2011)

Institute : Tokyo University of Agriculture & Technology
Research Area/Specialization : Protein solubilization, Sequence simplification Protein stabilization and Dengue sero-specificity.

Doctoral Research (October 2005-March 2009)

Institute : Tokyo University of Agriculture & Technology
Research Area/Specialization : Protein Sequence simplification and stabilization.

Graduation and Post-Graduation Research (August 1999-February 2003)

Institute :International Centre for Diarrheal Diseases Research, Bangladesh (ICDDR,B)
Research Area/Specialization :Immunological responses in human trial and in animal model.

Personal Research Grants (selected)

1. Research Theme : Institutional readiness for pandemic diseases……

   Research Period : 2025-2028
   Grant Authority : HEAT-UGC-WORLD Bank

2. Research Theme : Quantitative and qualitative detection of all natural dengue infections using a common recombinant dengue antigen

   Research Period : 2017-2019, 2021-2022
   Grant Authority : Ministry of Education, Bangladesh Government

3. Research Theme : Innovation of monoclonal antibodies against chronic diseases through university industry research collaboration

   Research Period : 2016-2018
   Grant Authority : World Bank and University Grant Commission, Bangladesh

4. Research Theme : Investigation of the effects of short peptide tags on the modulation of immunogenicity of a chimeric dengue ED3 in mice model

   Research Period : 2018-2019
   Grant Authority : Revenue Budget, Chittagong University

Research Collaboration

1. MoU : Memorandum of Understanding between Tokyo University of Agriculture and Technology and Chittagong University since 2014.
2. MTA : Material Transfer Agreement between Professor Yutaka Kuroda (of TUAT) and M. Monirul Islam (of CU) since 2013.
3. Student Exchange Program: JASSO International Student Exchange Program between TUAT and CU since 2015).
4. Lab-in-Charge : COVID-19 Lab, Chittagong University and Principle investigator: Anti-COVID-19 antibody responses in COVID-19 vaccinated people in Chittagong, Bangladesh.

List of Publications (Selected)

- "Biophysical and colloidal properties govern anti-drug antibody generation: A case study using the tumoricidal murine-derived monoclonal antibody 4713". Yesmin S, Islam MM, Islam MD, Tsurui H, Saotome T, Matsuoka S, Kuroda Y. Int J Biol Macromol. 2026. 355:151430
- Omicron RBD expressed in E. coli outperforms mammalian-expressed S1 spike protein in generating highly neutralizing anti-SARS-CoV-2 antibodies in mice. Zelin Wang, Rawiwan Wongnak, Mami Oba, Tetsuya Mizutani, M. Monirul Islam, Yutaka Kuroda, Virology, 619. 2026.
- Neutralizing antibodies against the Japanese encephalitis virus are produced by a 12 kDa E. coli- expressed envelope protein domain III (EDIII) tagged with a solubility-controlling peptide. Md. Din Islam, M. Monirul Islam, Ayae Inoue, Sanjida Yesmin, Subbaian Brindha, Takahiro Yoshizue, Hiromichi Tsurui, Takeshi Kurosu, Yutaka Kuroda, Vaccine, 56, 2025, 127143.
- Ni2+-induced selective precipitation of His-tagged recombinant proteins shortens purification time while maintaining high yield. Md Din Islam, M Monirul Islam, Yutaka Kuroda. Journal of Biotechnology. 399: 38-46. 2025.
- Non-Glycosylated SARS-CoV-2 Omicron BA.5 Receptor Binding Domain (RBD) with a Native-like Conformation Induces a Robust Immune Response with Potent Neutralization in a Mouse Model. Wongnak, R.; Brindha, S.; Oba, M.; Yoshizue, T.; Islam, M.D.; Islam, M.M.; Takemae, H.; Mizutani, T.; Kuroda, Y. *Molecules* **2024**, *29*, 2676.
- Toxins profiles, toxicological properties, and histological alteration potentiality of Trimeresurus erythrurus venom: In vitro and in vivo experiments. Ibrahim Khalil Al Haidar, Mohammad Abdul Wahed Chowdhury, Masum Miah, Mahedi Hasan, Sohanur Rohman, Mohammed Noman, Md Mizanur Rahman, Abdul Auawal, Sajib Rudra, Md Rafiqul Islam, Md Asir Uddin, Abdullah Abu Sayeed, Aniruddha Ghose, M Monirul Islam, Md Abu Reza. Journal of King Saud University-Science; Elsevier; page: 1-8; March **2024**.

- Anti-DENV-ED3 antibody cross-talks generate immune interference among the four DENVs. Md Din Islam, Sanjida Yesmin, Tahmina Sharmin, Md Ayoub Khan, Yutaka Kuroda, M Monirul Islam. Asian Pacific Journal of Tropical Medicine. 17: 553-562. 2024.
- Secondary DENV Infections Along with Concurrent Multiple DENV Serotypes in a Single Dengue Session Contributing to Dengue Severity in Chittagong Bangladesh. Moushumi Ghosh Roy , Kutub Uddin , Din Islam , M. Monirul Islam , Anjuvan Singh. Journal of Survey in Fisheries Sciences; https://doi.org/10.17762/sfs.v10i4S.845; page: 627-642; **2023**.
- The Immunogenicity of DENV1-4 ED3s Strongly Differ despite Their Almost Identical Three-Dimensional Structures and High Sequence Similarities. Md. Din Islam, Tahmina Sharmin, Imrul Hasan Tipo, Antara Saha, Sanjida Yesmin, Moushumi Ghosh Roy, Subbaian Brindha, Yutaka Kuroda, and M. Monirul Islam. https://doi.org/10.3390/ijms24032393; International Journal of Molecular Science; page: 24(3), 2393; **2023**,
- Systematic mutational analysis of epitope-grafted ED3 immunogenicity reveals a DENV3-DENV4 bi-serospecific ED3 mutant. Mamtaz Sultana, Nazmul Hasan, Mamunur R Mahib, Mohammad M Islam. Asian Pacific Journal of Tropical medince; DOI: 10.4103/1995-7645.338432; page: 15(2):63-70; March **2022**.
- Prevalence of dengue virus, their infection, diagnosis, and challenges. Moushumi Ghosh Roy, Mohammad Monirul Islam and Anjuvan Singh. Journal of Applied Biology & Biotechnology; page: 1-8; **2022**.
- All four dengue virus serotypes co-circulate in concurrent dengue infections in a single dengue session in Chittagong, Bangladesh. Moushumi Ghosh Roy, Kutub Uddin, Din Islam, Anjuvan Singh and Mohammad Monirul Islam. Bioresearch Communications; page: 8(1):1042-1048; October **2021**.
- Solubility Controlling Peptide Tags of Opposite Charges Generate a Bivalent Immune Response Against Dengue ED3 Serotypes 3 and 4. Nafsoon Rahman, Shiho Miura, Mami Okawa, Md. Golam Kibria, Mohammad Monirul Islam and Yutaka Kuroda. Front. Immunol.; Frontiers; June 2021.
- A systematic mutational analysis identifies a 5-residue proline tag that enhances the in vivo immunogenicity of a non-immunogenic model protein. Nafsoon Rahman Mohammad Monirul Islam Md. Golam Kibria Satoru Unzai Yutaka Kuroda. FEBS Openbio; FEBS Press; page: https://doi.org/10.1002/2211-5463.12941; July **2020**.
- Nanometer-size aggregates generated using short Solubility Controlling Peptide tags do increase the in vivo immunogenicity of a non-immunogenic protein. Rahman N, Islam MM, Unzai S, Miura S, Kuroda Y. Mol Pharm.; American Chemical Society (ACS); page: doi: 10.1021/acs.molpharmaceut.0c00071; March **2020.**
- Anti-Dengue ED3 long-term immune response with T-cell memory generated using Solubility Controlling Peptide tags. Islam MM, Miura S, Hasan MN, Rahman N, and Kuroda Y. Frontiers in Immunology Vaccines and Molecular Therapeutics; page: Volume 11; Article. 333; March **2020**.
- Hydrophobic surface residues can stabilize a protein through improved water-protein interactions. Islam MM, Kobayashi K, Kidokoro SI, Kuroda Y. FEBS Journal; FEBS Press; page: doi: 10.1111/febs.14941.; June **2019**.
- Reversible association of proteins into sub-visible amorphous aggregates using short solubility controlling peptide tags. MG Kabir, MM Islam, Y Kuroda. Biochimica et Biophysica Acta (BBA)-Proteins and Proteomics; page: 1866 (2), 366-372; **2018**.
- A hetero-micro-seeding strategy for readily crystallizing closely related protein variants. MM Islam, Y Kuroda. Biochemical and biophysical research communications; page: 493 (1), 504-508; **2017**.

- Crystal structures of highly simplified BPTIs provide insights into hydration-driven increase of unfolding enthalpy. MM Islam, M Yohda, S Kidokoro, Y Kuroda. Scientific reports; page: 7, 41205; **2017**.
- Unusual reversible oligomerization of unfolded dengue envelope protein domain 3 at high temperatures and its abolition by a point mutation. T Saotome, S Nakamura, MM Islam, A Nakazawa, M Dellarole, F Arisaka,... Biochemistry 55 (32), 4469-4475, 2016; **2016**.
- Analysis and control of protein crystallization using short peptide tags that change solubility without affecting structure, thermal stability and function. Islam MM, Nakamura S, Noguchi K, Yohda M, Kidokoro SI, and Kuroda Y. J. Crystal Growth and Design,; page: 15 (6), 2703-2711; 2015.
- Structural and biophysical analysis of sero-specific immune responses using epitope grafted Dengue ED3 mutants. Kulkarni MR, Islam MM, Mahib MR, Numoto N, Elahi M, Ito N, and Kuroda Y. Biochim. Biophys. Acta (BBA)-Proteins and Proteomics,; page: 854 (10), 1438-1443; **2015**.
- Computational prediction and experimental characterization of a "size switch type repacking" during the evolution of dengue envelope protein domain III (ED3). Elahi M, Islam MM, Noguchi K, Yohda M, Toh H, and Kuroda Y. Research Article; Biochim. Biophys. Acta.; page: 1844(3):6585-92.; **2014**.
- High resolution crystal structure of dengue-3 envelope protein domain III suggests possible molecular mechanisms for serospecific antibody recognition. Elahi M, Islam MM, Noguchi K, Yohda M, and Kuroda Y. Research Article; PROTEINS; Structure, Function and Bioinformatics; **2013**.
- Analysis of protein aggregation kinetics using short amino acid peptide tags. Khan MA, Islam MM, and Kuroda Y. Research Article; Biochim. Biophys. Acta.; page: 1834(10): 2107-15.; **2013**.
- Probing the Physical Determinants of Thermal Expansion of Folded Proteins. Dellarole M, Kobayashi K, Rouget JP, Caro A, Roche J, Islam MM, Moreno BG, Kuroda Y, and Royer CA. Research Article; J. of Phys. Chem.; page: B117(42):12742-9.; **2013**.
- Analysis of amino acid contributions to protein solubility using short peptide tags fused to a simplified BPTI variant. Islam MM, Khan MA, and Kuroda Y. Research Article; Biochim. Biophys. Acta.; page: 1824(10): 1140-50.; **2012**.
- Information needed to specify a protein structure: structure and thermodynamics of highly simplified bovine pancreatic trypsin inhibitor. Kato A, Islam MM, and Kuroda Y. Research Article; Tanpakushitsu Kakusan Koso,; page: 54(5):643-648.; **2009**.
- Thermodynamic and structural analysis of highly stabilized BPTIs by a single and double mutations. Islam MM, Sohya S, Noguchi K, Kidokoro SI, Yohda M, and Kuroda Y. Research Article; PROTEINS; Structure, Function and Bioinformatics,; page: 77(4):962-970.; **2009**.
- Crystal structure of an extensively simplified bovine pancreatic trypsin inhibitor in which over one-third of the residues are alanine. Islam MM, Sohya S, Noguchi K, Yohda M, and Kuroda Y. Research Article; Proc. Natl. Acad. Sci. USA,; page: 105:15534-15539.; **2008.**
- Dyslipidemia in Type II Diabetes and the effect of Glycaemic control on Type II Diabetes. Banik GR, Islam MM, Zafar MA. Research Article; Hamdard medicus; page: 49 (4), 55-62; **2006**.
- Detection of Vibrio cholerae O1 and O139 serogroups directly from stool specimens by combined immunomagnetic separation and polymerase chain reaction. MM Islam, MG Kabir, PK Moly, A Islam, MA Rakib,. Research Article; Pakistan J Biol Sci; page: 7, 1654-9; **2004**.
- Coagglutination: A rapid and sensitive assay method for detectionof Vibrio cholerate O1 and O139 serogroups directly from stool specimens. MM Islam, MG Kabir, AFMT Islam, Research Article; Pakistan Journal of Biological Sciences; **2004**.

- “Antigen-specific immunoglobulin A antibodies secreted from circulating B cells are an effective marker for recent local immune responses in patients with cholera: comparison to antibody-secreting cell responses and other immunological markers”. Qadri F, Ryan ET, Faruque AGS, Ahmed F, Khan AI, Islam MM, Akramuzzaman SM, Sack DA, Calderwood SB. Research Article; Infect. Immun.,; page: 71(8):4808-4814.; **2003**.